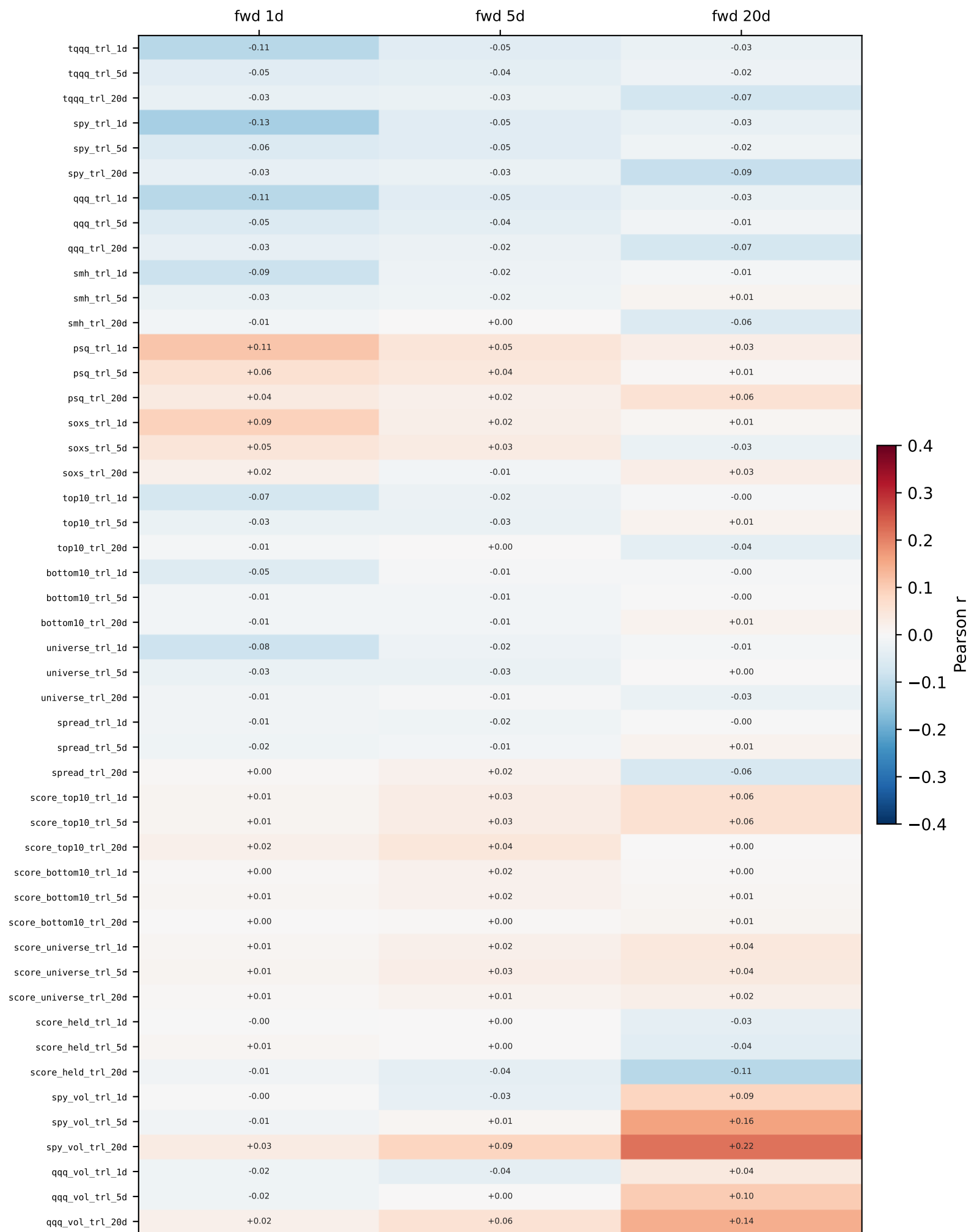


TQQQ — trailing model_v4 metric → forward TQQQ return (full-sample corr, 2514 days 2016-2026)



TQQQ — tail call-outs: which model_v4 trailing metrics separate the forward-TQQQ

UP = forward TQQQ return > +thr, DN < -thr (thr = 60th pct of |fwd|). D = UP avg - DN avg of the trailing metric; r = its in-tail corr with the f

FWD 1D (thr ±2.6%, UP n=562 / DN n=443) — top metrics by |in-tail corr|

spy_vol_trl_5d	UP r=+0.48 DN r=-0.44	D= -0.011	(UP +0.169 / DN +0.180)
qqq_vol_trl_5d	UP r=+0.45 DN r=-0.40	D= -0.016	(UP +0.222 / DN +0.237)
spy_vol_trl_1d	UP r=+0.37 DN r=-0.39	D= -0.006	(UP +0.138 / DN +0.144)
spy_trl_20d	UP r=-0.46 DN r=+0.33	D= +0.0%	(UP +0.3% / DN +0.3%)
spy_vol_trl_20d	UP r=+0.46 DN r=-0.32	D= -0.002	(UP +0.179 / DN +0.181)
qqq_vol_trl_20d	UP r=+0.47 DN r=-0.32	D= -0.006	(UP +0.231 / DN +0.237)
qqq_vol_trl_1d	UP r=+0.31 DN r=-0.31	D= -0.015	(UP +0.177 / DN +0.192)

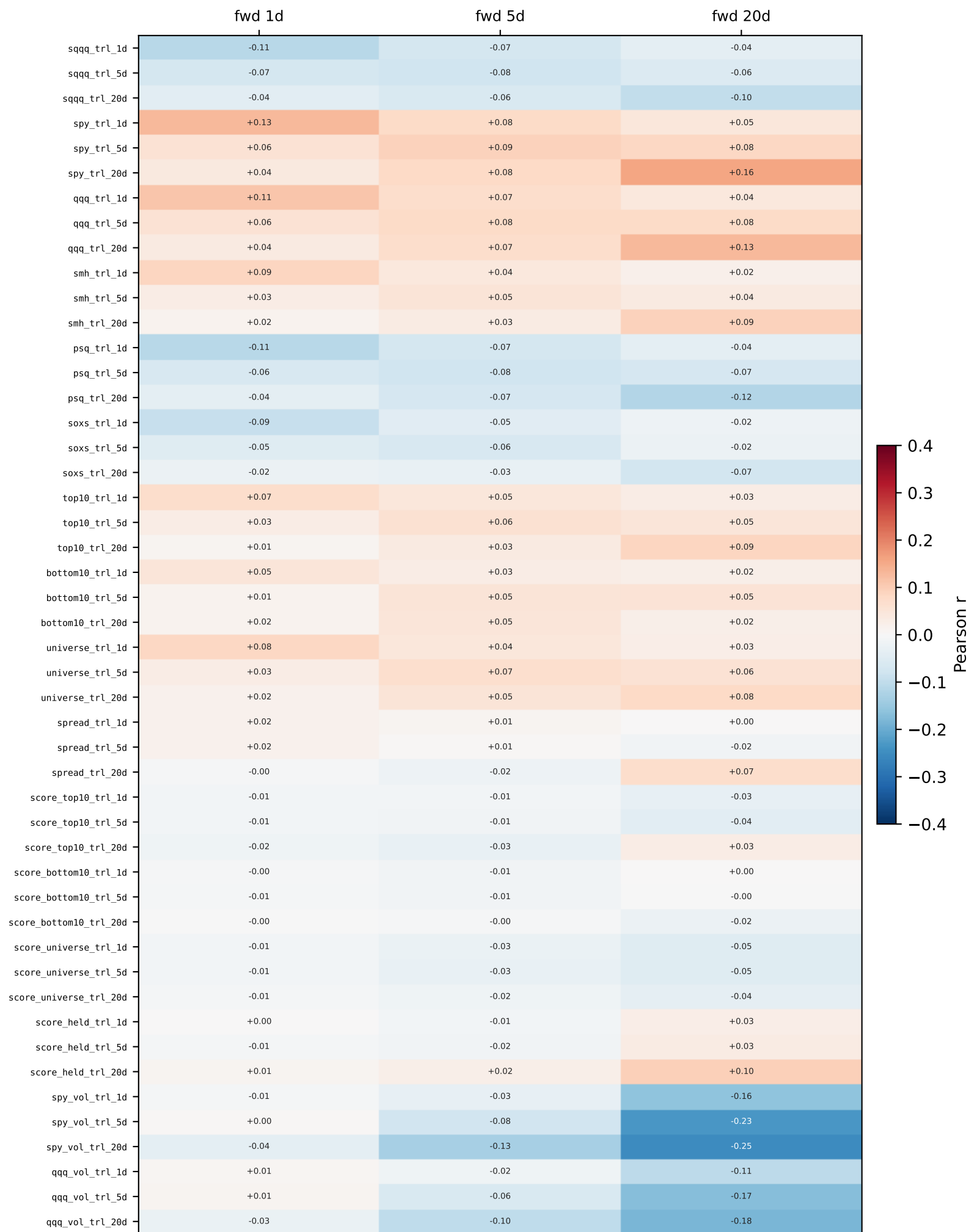
FWD 5D (thr ±6.3%, UP n=589 / DN n=415) — top metrics by |in-tail corr|

spy_vol_trl_5d	UP r=+0.40 DN r=-0.38	D= -0.002	(UP +0.172 / DN +0.174)
qqq_vol_trl_5d	UP r=+0.42 DN r=-0.35	D= -0.010	(UP +0.222 / DN +0.233)
spy_vol_trl_20d	UP r=+0.47 DN r=-0.35	D= +0.014	(UP +0.181 / DN +0.167)
spy_vol_trl_1d	UP r=+0.28 DN r=-0.34	D= -0.010	(UP +0.140 / DN +0.150)
qqq_vol_trl_20d	UP r=+0.50 DN r=-0.33	D= +0.003	(UP +0.231 / DN +0.228)
qqq_vol_trl_1d	UP r=+0.26 DN r=-0.30	D= -0.017	(UP +0.181 / DN +0.198)
spy_trl_20d	UP r=-0.32 DN r=+0.25	D= +0.0%	(UP +0.4% / DN +0.4%)

FWD 20D (thr ±14.1%, UP n=660 / DN n=338) — top metrics by |in-tail corr|

score_held_trl_1d	UP r=+0.18 DN r=-0.20	D= -0.011	(UP +0.594 / DN +0.605)
smh_trl_20d	UP r=-0.31 DN r=+0.17	D= +0.3%	(UP +1.0% / DN +0.7%)
spy_vol_trl_5d	UP r=+0.50 DN r=-0.16	D= +0.018	(UP +0.182 / DN +0.164)
spy_vol_trl_20d	UP r=+0.52 DN r=-0.15	D= +0.030	(UP +0.196 / DN +0.166)
soxs_trl_20d	UP r=+0.19 DN r=-0.15	D= -1.1%	(UP -5.0% / DN -3.8%)
qqq_vol_trl_5d	UP r=+0.48 DN r=-0.14	D= -0.001	(UP +0.224 / DN +0.224)
spy_vol_trl_1d	UP r=+0.35 DN r=-0.14	D= +0.006	(UP +0.147 / DN +0.141)

SQQQ — trailing model_v4 metric → forward SQQQ return (full-sample corr, 2514 days 2016-2026)



SQQQ — tail call-outs: which model_v4 trailing metrics separate the forward-SQQQ

UP = forward SQQQ return > +thr, DN < -thr (thr = 60th pct of |fwd|). D = UP avg - DN avg of the trailing metric; r = its in-tail corr with the f

FWD 1D (thr ±2.6%, UP n=441 / DN n=564) — top metrics by |in-tail corr|

spy_vol_trl_5d	UP r=+0.42	DN r=-0.49	D= +0.011	(UP +0.180 / DN +0.169)
qqq_vol_trl_20d	UP r=+0.32	DN r=-0.47	D= +0.005	(UP +0.236 / DN +0.230)
spy_vol_trl_20d	UP r=+0.33	DN r=-0.46	D= +0.001	(UP +0.179 / DN +0.178)
qqq_vol_trl_5d	UP r=+0.38	DN r=-0.46	D= +0.015	(UP +0.237 / DN +0.222)
spy_trl_20d	UP r=-0.32	DN r=+0.46	D= -0.0%	(UP +0.3% / DN +0.3%)
qqq_trl_20d	UP r=-0.29	DN r=+0.41	D= +0.0%	(UP +0.5% / DN +0.5%)
psq_trl_20d	UP r=+0.27	DN r=-0.40	D= -0.0%	(UP -0.4% / DN -0.4%)

FWD 5D (thr ±6.2%, UP n=392 / DN n=612) — top metrics by |in-tail corr|

qqq_vol_trl_20d	UP r=+0.25	DN r=-0.51	D= -0.016	(UP +0.222 / DN +0.238)
spy_vol_trl_20d	UP r=+0.24	DN r=-0.49	D= -0.025	(UP +0.162 / DN +0.187)
qqq_vol_trl_5d	UP r=+0.22	DN r=-0.44	D= -0.008	(UP +0.224 / DN +0.232)
spy_vol_trl_5d	UP r=+0.21	DN r=-0.43	D= -0.014	(UP +0.166 / DN +0.180)
spy_trl_20d	UP r=-0.12	DN r=+0.36	D= +0.5%	(UP +0.7% / DN +0.1%)
qqq_trl_20d	UP r=-0.11	DN r=+0.32	D= +0.6%	(UP +0.9% / DN +0.3%)
universe_trl_20d	UP r=-0.09	DN r=+0.30	D= +0.4%	(UP +1.4% / DN +1.0%)

FWD 20D (thr ±13.3%, UP n=278 / DN n=720) — top metrics by |in-tail corr|

spy_vol_trl_5d	UP r=-0.01	DN r=-0.60	D= -0.043	(UP +0.147 / DN +0.190)
spy_vol_trl_20d	UP r=+0.08	DN r=-0.59	D= -0.048	(UP +0.155 / DN +0.203)
qqq_vol_trl_20d	UP r=+0.08	DN r=-0.59	D= -0.028	(UP +0.220 / DN +0.248)
qqq_vol_trl_5d	UP r=-0.00	DN r=-0.57	D= -0.028	(UP +0.206 / DN +0.234)
spy_trl_20d	UP r=+0.11	DN r=+0.51	D= +1.0%	(UP +1.4% / DN +0.4%)
universe_trl_20d	UP r=+0.05	DN r=+0.45	D= +0.5%	(UP +2.1% / DN +1.6%)
spy_vol_trl_1d	UP r=+0.02	DN r=-0.44	D= -0.034	(UP +0.123 / DN +0.157)

TQQQ vs SQQQ — strongest trailing→forward predictors (full-sample |r|)

TQQQ:

spy_vol_trl_20d	fwd_20d	r = +0.22
spy_vol_trl_5d	fwd_20d	r = +0.16
qqq_vol_trl_20d	fwd_20d	r = +0.14
spy_trl_1d	fwd_1d	r = -0.13
qqq_trl_1d	fwd_1d	r = -0.11
psq_trl_1d	fwd_1d	r = +0.11
tqqq_trl_1d	fwd_1d	r = -0.11
score_held_trl_20d	fwd_20d	r = -0.11
qqq_vol_trl_5d	fwd_20d	r = +0.10
soxs_trl_1d	fwd_1d	r = +0.09
spy_trl_20d	fwd_20d	r = -0.09
smh_trl_1d	fwd_1d	r = -0.09

SQQQ:

spy_vol_trl_20d	fwd_20d	r = -0.25
spy_vol_trl_5d	fwd_20d	r = -0.23
qqq_vol_trl_20d	fwd_20d	r = -0.18
qqq_vol_trl_5d	fwd_20d	r = -0.17
spy_vol_trl_1d	fwd_20d	r = -0.16
spy_trl_20d	fwd_20d	r = +0.16
spy_vol_trl_20d	fwd_5d	r = -0.13
spy_trl_1d	fwd_1d	r = +0.13
qqq_trl_20d	fwd_20d	r = +0.13
psq_trl_20d	fwd_20d	r = -0.12
sqqq_trl_1d	fwd_1d	r = -0.11
qqq_trl_1d	fwd_1d	r = +0.11